

8	(a)	(i)	To remain at rest, net force = 0. $W = QE$ or at minimum $W = Q \left(\frac{V}{d} \right)$ leading to $Q = W \left(\frac{d}{V} \right)$ B1	[1]
		(ii)	electric force upwards M1 (to balance downward weight or to suspend oil droplets) hence A is at a higher potential A1 (for downward electric field or to attract negative charges)	[2]

		(iii)	density of air $\ll$ density of oil	B1 [1]
	(b)		Correct W : 5.1 or 5.2 (read off from graph in Fig. 7.2 at $T = 7.6$ s) Correct Q (allow ecf): 4.80 (for $W = 5.1$) or 4.89 (for $W = 5.2$) or $Q = W(d/V)$ calculated correctly for wrong values of W stated	B1 B1 [2]
	(c)		At terminal speed, net force = 0 Substitution of $F_D (= W) = 2.9 \times 10^{-14}$ N (accept substitution of $W = Q(V/d) = 1.66 \times 10^{-19}$ (770 / 4.42 $\times 10^{-3}$) even though it is not necessary) Substitution of $v (= D/T) = 1.00 \times 10^{-3} / 11.2$ (allow this mark for erroneous substitution of D e.g. 1.00, 4.42 or 4.42 $\times 10^{-3}$) (do not allow for any substitution of v using any equation of motion where $a \neq 0$) $r = 9.57 \times 10^{-7}$ m (accept 9.58 $\times 10^{-7}$ due to early round-off)	C1 C1 A1 [3]
	(d)	(i)	From Fig. 7.2, When $T = 8.0$ s, $W = 4.7$ or 4.8×10^{-14} N, $\Rightarrow \lg W = -13.33$ or -13.32 (2dp)	B1 [1]
		(ii)	correct point plotted within half a square best fit line drawn	B1 B1 [2]
		(iii)	$W = aT^b \Rightarrow \lg W = b \lg T + \lg a$ $b = \text{gradient}$ $= \frac{(-13.14) - (-13.86)}{0.78 - 1.26} = \frac{0.72}{-0.48} = -1.5$ e.g. an attempt to calculate gradient or clear working that $b = \text{gradient}$ gradient calculated correctly with points chosen on best fit line at least half the length of the drawn best fit line apart	C1 A1 [2]
		(iv)	$W = aT^{-1.5} = a\left(\frac{1}{T}\right)^{1.5}$ Terminal speed $v = \frac{d}{T}$ $\Rightarrow W = a\left(\frac{v}{d}\right)^{1.5} \propto v^{1.5}$ since a and d are constants Substitution of the value of b into $W = aT^b$ or $\lg W = b \lg T + \lg a$ or $v = D/T$ leading to $W = \frac{a}{D^{1.5}} v^{1.5}$ or $W \propto v^{1.5}$	C1 A1 [2]
	(e)		any electric charge is always an integer multiple of a basic charge or there is a certain basic charge beyond which the charge is no longer divisible into smaller units	B1 [1]

	(f)	at least one value of Q divided by another value of Q or at least one difference between the values of Q determined C1 at least 5 values of Q or differences between values of Q divided by the smallest value of Q or smallest difference between values of Q or divide 5 other values of Q by $Q_5/4$ C1 e.g. $Q_1 = N_1q$ $Q_2 = N_2q$ $Q_3 = N_3q$ $Q_4 = N_4q$ $Q_5 = N_5q$ $Q_6 = N_6q$ largest possible q corresponds to smallest possible N_N (choose to check Q_5 and N_5 since smallest N should be easiest to work with) $Q_1/Q_5 = N_1/N_5 = 7/2$ Since N_1 must be a positive integer, $N_5 = 2n$ where n is a positive integer $Q_2/Q_5 = N_2/N_5 = 9/4$ $N_5 = 4n$ $Q_3/Q_5 = N_3/N_5 = 17/4$ $N_5 = 4n$ $Q_4/Q_5 = N_4/N_5 = 11/4$ $N_5 = 4n$ $Q_6/Q_5 = N_6/N_5 = 11/4$ $N_5 = 2n$ smallest possible $N_5 = 4$ OR <table><tr><td></td><td colspan="6">Q / 10⁻⁹ units</td><td></td></tr><tr><td></td><td>6.86</td><td>4.44</td><td>8.37</td><td>5.39</td><td>1.97</td><td>2.96</td><td></td></tr><tr><td>+1.97</td><td>3.48</td><td>2.25</td><td>4.25</td><td>2.74</td><td>1.00</td><td>1.50</td><td></td></tr><tr><td>+(1.97/2)</td><td>6.96</td><td>4.51</td><td>8.50</td><td>5.47</td><td>2.00</td><td>3.01</td><td></td></tr><tr><td>+(1.97/3)</td><td>10.45</td><td>6.76</td><td>12.75</td><td>8.21</td><td>3.00</td><td>4.51</td><td></td></tr><tr><td>+(1.97/4)</td><td>13.93</td><td>9.02</td><td>16.99</td><td>10.94</td><td>4.00</td><td>6.01</td><td>← all are close to whole number</td></tr></table> therefore largest possible $q = Q_5 / 4 = 1.97 \times 10^{-9} \div 4 = 4.93 \times 10^{-10}$ units A1		Q / 10 ⁻⁹ units								6.86	4.44	8.37	5.39	1.97	2.96		+1.97	3.48	2.25	4.25	2.74	1.00	1.50		+(1.97/2)	6.96	4.51	8.50	5.47	2.00	3.01		+(1.97/3)	10.45	6.76	12.75	8.21	3.00	4.51		+(1.97/4)	13.93	9.02	16.99	10.94	4.00	6.01	← all are close to whole number	[3]
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